ChronoFuse at SemEval-2027 Task 1: Metadata- and Time-Aware Hybrid Retrieval with Guidance-Distilled Temporal Query Understanding

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Abstract

Temporal reasoning-intensive retrieval requires finding passages that are not only topically relevant but also aligned with the time periods a question depends on, a setting where lexical retrievers perform poorly (BM25 reaches only 0.115 nDCG@10 on the RETECO Track 1 development set). We present ChronoFuse, our system for Sub-tracks 1a and 1b of SemEval-2027 Task 1 (RETECO). ChronoFuse (i) extracts temporal and entity metadata from every document and stores it alongside sparse, dense and knowledge-graph indexes; (ii) analyses each query with an LLM prompted by *guidance-distilled* demonstrations that reproduce the benchmark's temporal annotation schema (intent, anchors, events, year window, retrieval plan); (iii) retrieves through six complementary channels fused with weighted reciprocal rank fusion and a temporal-alignment prior; and (iv) reranks with a metadata-augmented cross-encoder followed by a temporal-aware listwise LLM. ChronoFuse reaches [TODO: X] nDCG@10 on 1a and [TODO: Y] on 1b (dev), improving over BM25 by [TODO: Z] points; ablations show [TODO: key finding]. Code is available at [TODO: URL].

1 Introduction

Retrieval-augmented generation (RAG) systems assume that the retriever surfaces evidence that is *valid for the question being asked*. For temporally grounded questions—"Why was there a jump in transactions per block in 2023?"—validity depends on time: a passage about the right topic but the wrong period is misleading evidence. The RETECO shared task (Abdallah et al., 2026; Ali et al., 2026) operationalises this with TEMPO, a 13-domain benchmark of StackExchange questions whose relevant passages must be both topical and temporally aligned.

The official BM25 baseline reaches only 0.108–0.118 nDCG@10 on Track 1, far below its 0.44 on the conversational Track 2, confirming that "temporal grounding is not a keyword problem". We hypothesise three reasons: (1) queries often express time *implicitly* ("after the halving", "two weeks ago"), (2) relevant passages frequently share few terms with the question, and (3) nothing in a bag-of-words ranking prefers documents from the right period.

ChronoFuse addresses each: [TODO: one sentence per component, mapped to (1)–(3)].

Our contributions are:

- A metadata layer that attaches normalised temporal profiles, entities and short descriptions to 1.17M documents and exposes them to every retrieval and reranking stage.
- *Guidance distillation*: using TEMPO's train-split query guidance as demonstrations so an LLM emits the benchmark's own temporal annotation schema for unseen queries.
- A six-channel hybrid retriever (sparse, dense, HyDE, sub-query, temporal knowledge graph) with weighted RRF and a temporal-alignment prior, plus a temporal-aware listwise reranker.
- Ablations and per-domain / per-intent analyses on all 13 domains.

2 Background

Task. Track 1 of RETECO (TEMPO) contains 1,730 train/dev queries over 13 domains (blockchain, social sciences, applied and STEM StackExchange sites), each with its own corpus (1,167,159 documents in total, v1.1). In Sub-track 1a a system ranks documents for the whole query; in Sub-track 1b it ranks documents for each of 3,976 decomposed reasoning steps. The official metric is binary nDCG@10, averaged per domain and then macro-averaged across domains.

Train/dev splits are 70/30 per domain; the hidden test set has ∼350 newly annotated queries.

Guidance annotations. For train/dev queries, TEMPO additionally releases *query guidance*: whether the query is temporal, its temporal intent, expected granularity, time anchors, temporal events and a step-wise retrieval plan, plus per-passage temporal annotations. These are unavailable at test time; CHRONOFUSE uses them only as supervision.

Related work. *Hybrid retrieval and fusion:* BM25 (Robertson and Zaragoza, 2009), dense bi-encoders (Karpukhin et al., 2020; Xiao et al., 2023), and reciprocal rank fusion (Cormack et al., 2009). *LLM-based query expansion and reranking:* HyDE (Gao et al., 2023) and listwise LLM reranking (Sun et al., 2023). *Structure-aware RAG:* hierarchical summarisation (Sarthi et al., 2024) and graph-based retrieval (Edge et al., 2024; Gutiérrez et al., 2024). *Temporal IR:* temporal expression tagging (Strötgen and Gertz, 2010) and time-aware ranking (Campos et al., 2014). *Reasoning-intensive retrieval:* BRIGHT (Su et al., 2025). [TODO: position CHRONOFUSE relative to TEMPO's reported best systems]

3 System Overview

Figure 1 summarises CHRONOFUSE. All stages operate per domain.

3.1 Document Processing and Metadata

Documents are split into 200-word passages with 40-word overlap (at most 64 per document); document scores are aggregated with MaxP. For each document we extract a *temporal profile*: explicit years, ranges, month–year and ISO dates, decades and centuries, normalised to a set of years with a granularity label, and relative expressions ("two weeks ago", "no longer"). We also store a proxy title, an extractive short description and salient entities. [TODO: coverage statistics: % docs with ≥1 year, per domain]

3.2 Indexes

Sparse: Lucene-variant BM25 (k_1=0.9, b=0.4) over passages. **Dense**: [TODO: encoder] embeddings in a vector database (Qdrant) whose payload carries the temporal profile, enabling year-window filtering. **Temporal knowledge graph**: a graph with document, entity and year nodes and MENTIONS, REFERSTO and ACTIVEIN edges; retrieval uses personalised PageRank seeded with the query's entities and years.

3.3 Guidance-Distilled Query Analysis

An LLM receives n train-split (query, guidance) pairs as demonstrations and outputs, for a new query, the TEMPO schema (temporal intent, granularity, anchors, events, retrieval plan) extended with retrieval fields: an inferred year window, entities, a keyword query and a HyDE passage. [TODO: analyzer accuracy vs. gold guidance on dev: intent accuracy, window overlap]

3.4 Multi-Channel Retrieval and Fusion

Six channels are fused with weighted RRF, $s(d) = \sum_c w_c/(k + \mathrm{rank}_c(d))$, k=60: BM25 on the raw query, BM25 on the keyword query, RRF over BM25 runs of the sub-queries, dense on the query, dense on the HyDE passage, and KG-PPR.

3.5 Temporal-Alignment Prior

The fused score is interpolated with a temporal alignment term,

$$s'(d) = (1 - \lambda)\,\widehat{s}(d) + \lambda\,T(q, d), \qquad (1)$$

where $\widehat{s}$ is min–max normalised and $T(q, d)$ is the fraction of the query's window years matched by the document's years (±1 year). λ is tuned on train.

3.6 Reranking

A cross-encoder ([TODO: model]) scores the top-100 candidates rendered as "[years] title — passage". Finally, an LLM reranks the top-40 with a RankGPT-style sliding window; its prompt contains the analysed temporal need and instructs the model to prefer temporally aligned evidence.

3.7 Sub-track 1b

For each step, the query is the parent question's title concatenated with the step instruction; the parent query's analysis (window, entities) is shared across its steps.

4 Experimental Setup

Data and protocol. We use RETECO v1.1. Hyperparameters (λ, channel weights, number of demonstrations) are tuned on the train split; we report dev results once per configuration. Scoring follows the official protocol: `pytrec_eval`

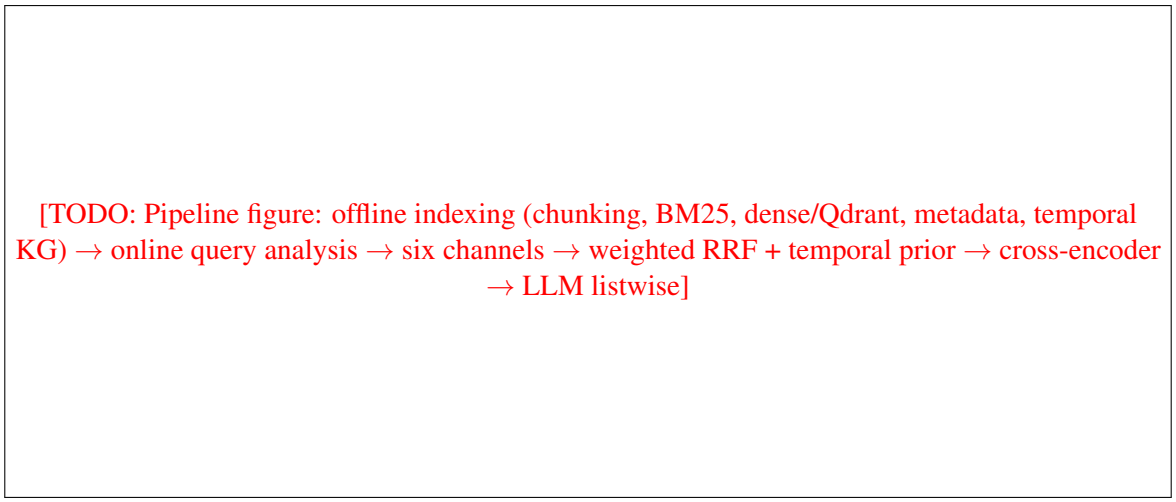

Figure 1: Overview of ChronoFuse.

	1a		1b	
System	train	dev	train	dev
BM25 (official)	.108	.115	.102	.118
E0 BM25 passages+MaxP	–	–	–	–
E1 + temporal prior	–	–	–	–
E2 + dense + KG (RRF)	–	–	–	–
E3 ChronoFuse (full)	–	–	–	–

Table 1: Macro nDCG@10 on RETECO Track 1 (v1.1). Official baseline from the starter kit.

`ndcg_cut_10`, per-domain mean, macro-average over 13 domains; topics without a ranking count as zero.

Models. [TODO: dense encoder, cross-encoder, LLM (name + version), API vs. local]

Compute. [TODO: hardware, index build time, per-query latency, LLM cost]

5 Results

Main results. [TODO: Table 1]

Ablations. [TODO: remove one channel / component at a time from E3]

Per-domain analysis. [TODO: Figure: per-domain nDCG@10, BM25 vs. ChronoFuse]

Temporal intent analysis. [TODO: nDCG@10 grouped by gold temporal intent / reasoning class from guidance; temporal diagnostics TP@10, TR@10, TC@10]

Error analysis. [TODO: qualitative examples]

6 Conclusion

[TODO: summary of findings and future work]

Limitations

Rule-based temporal extraction misses implicit periods expressed through events, which the LLM analyzer only partially recovers. LLM-based components add cost and latency and depend on a specific model version. All data is English StackExchange text, so findings may not transfer to other languages or genres. [TODO: expand]

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A Prompts

[TODO: query-analysis prompt and listwise reranking prompt]

B Hyperparameters

[TODO: full configuration table (configs/*.yaml)]